

University of Management and Technology Lahore
School of Science and Technology
Department of Computer Science

Complex Computing Problem (CCP)

Activity Selection Problem

Semester: Spring 2026
Course: CS3044 Analysis of Algorithms
Marks: 7

Submission Deadline: 13 July, 2026

TABLE I
COURSE LEARNING OUTCOMES (CLOs).

CLOs	Description
CLO 1	ARGUE and PROVE the correctness of algorithms using rigorous mathematical techniques taught in this course. (Level: C4)
CLO 2	ANALYZE the time complexity of different algorithms. (Level: C3)
CLO 3	COMPARATIVE analysis of different algorithm design techniques. (Level: C4)
CLO 4	DESIGN algorithms for different kinds of problems using efficient algorithm design techniques for real life problems. (Level: C4)
CLO 5	Propose a solution to a given design problem while adhering to professional ethics and norms of computing practices. There must be a debate among team members before every design decision and students must appreciate ethical problems related to teamwork and plagiarism (A3)
CLO 6	Develop a report on the entire process of designing the solution of given CCP such that every step is communicated effectively. The students should mention in the report the argument through which they compared the design choices to get to the final decision. The student will also present orally the results and process of the design at the end of the entire activity. (A4)



TABLE II
PROGRAM LEARNING OUTCOMES (PLOs) FOR CCP.

PLO	Description
2	Knowledge for Solving Computing Problems: Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Problem Analysis: Identify and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and Development of Solutions: Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

TABLE III
Range of Complex Problem Solving

CCP Attribute	Attribute	A Complex Computing Problem
1	Range of conflicting requirements	Involves wide-ranging or conflicting technical, computing, and other issues.
2	Depth of analysis required	Has no obvious solution, and requires conceptual thinking and innovative analysis to formulate suitable abstract models?
3	Depth of knowledge required	A solution requires the use of in-depth computing or domain knowledge and an analytical approach that is based on well-founded principles.

TABLE IV
Blooms Taxonomy Domain Levels

CCP Attribute.	Domain	Description	Bloom's Taxonomy Level
1	Cognitive	Understanding: Grasp the meaning of materials	C2
2	Cognitive	Applying: Use the information in a new situation.	C3



3	Cognitive	Analyzing: Identify schemas or relationships.	C4
3	Affective	Valuing: Attach values and express personal opinion	A3



TABLE IV
ASSESSMENT RUBRIC AND DELIVERABLES (MAX. MARKS:15).

Assessment Criteria	CCP Attribute	CLO / PLO	Outstanding	Effective	Inadequate
Problem analysis results in constraints to be imposed on the solution.	Range of conflicting requirements	2/3	2	2	1
Design including Flow chart / Block diagrams, and comparison if alternative solutions.	Depth of knowledge required/ Depth of analysis required	5/4	3	2	1
Report originality		6/4	2	1	0.5
Viva.	Depth of knowledge required, Depth of analysis required	6/4	3	2	1



Complex Computing Problem Statement

Design and analyze the solution for the activity selection problem using **brute force, dynamic programming, and greedy approach**. The Activity Selection Problem is an optimization problem that deals with selecting non-conflicting activities that need to be executed by a single person or machine in a given time frame. Let's consider that you have n activities with their start and finish times, the objective is to find the solution set having a **maximum number of non-conflicting activities** that can be executed in a single time frame, assuming that only one person or machine is available for execution.

- It might not be possible to complete all the activities since their timings can collapse.
- Two activities, say i and j , are said to be non-conflicting if $s_i \geq f_j$ or $s_j \geq f_i$ where s_i and s_j denote the starting time of activities i and j respectively, and f_i and f_j refer to the finishing time of the activities i and j respectively.

Input Data for the Algorithm:

- $act[]$ array containing all the activities.
- $s[]$ array containing the starting time of all the activities.
- $f[]$ array containing the finishing time of all the activities.

Output Data from the Algorithm:

- $sol[]$ array referring to the solution set containing the maximum number of non-conflicting activities.

The solution provided must satisfy the following

constraints **Range of conflicting requirements**

The assigned problem contains multiple conflicting activities in terms of the start and finish time. The student needs to design a solution by avoiding all the conflicts and selecting the maximum number of non-conflicting activities that can be performed.

Depth of knowledge required

The student should research the various algorithm design techniques such as brute force, dynamic programming, and greedy. They also need to understand how a problem can be broken down into smaller problems and how the solution of smaller problems can be used to find the solution of bigger size problems.

Depth of analysis required

A detailed analysis of all the proposed solutions is also required to find out the best approach that can be used to solve the activity selection problem.



Report submission tasks:

- 1) The report must be written in 12-point font size with Times New Roman font style.
- 2) The report must contain details of the algorithms/techniques used, with proper captions for the algorithm, figures and tables, and graphs for results. Make your figures in MS VISIO 2019/ Adobe Photoshop or any other software
- 3) Plagiarism must be less than 10%. Your report will be uploaded on Turnitin.com for plagiarism check. Write in your own words.

The following should be added in the technical report.

1. Title Page
2. Abstract/ Summary not more than 250 words with Keywords
3. Introduction of the activity selection problem.
4. Design and analyze the brute force solution for the activity selection problem.
5. Design and analyze the dynamic programming solution for the activity selection problem.
6. Design and analyze the greedy solution for the activity selection problem.
7. Comparison of used approaches
8. Conclusion

